

Grade 6 Unit 9 Assessment Guide

General Information	Unit 9 introduces students to finding and interpreting measures of central tendency and measures of variability. Students also learn how to read box plots and histograms in preparation for the students actually creating the box plots or histograms. This assessment offers a balanced set of questions where the student can show their knowledge of the concepts learned in Unit 9. To help teachers determine if students struggle distinguishing between the mean and median, these measures of center were not asked within the same context. Wherever box plots and histograms are provided, the student also has the data so that the student does not just have to read the data from a diagram. Students are not assessed on comparing data sets as these lessons were to help the student solidify their understanding of the displays, all of which will support student learning in Unit 10. As these concepts are new to the student, it may be helpful for the teacher to monitor student understanding using this assessment in preparation for Unit 10, where students create displays thus relying upon their learning in Unit 9. If a teacher notes an overall weakness, then the teacher may wish to reteach the concept before Unit 10 or take opportunities within the lessons of Unit 10 to reteach. Students are allowed the use of a calculator for this assessment because the focus should be on understanding how to find each measure. Most of the questions on this assessment are not calculator dependent. Teachers should note that an assessment limit in the draft item specifications states that items that assess finding the mean are restricted to 10 values. This restriction is likely in place so that difficulty is easier so that students could do the mathematics without a calculator. Teachers could easily reorder the questions to create a two-part assessment where one part is calculator and one part is not calculator, if preferred.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.6.DP.1.2
38		Recommended Scoring (5 Total Points)	Consider not giving partial credit.
Question	1b	Benchmark(s)	MA.6.DP.1.2
22		Recommended Scoring (5 Total Points)	Consider not giving partial credit.
Question	1c	Benchmark(s)	MA.6.DP.1.2
30		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	2a	Benchmark(s)	MA.6.DP.1.3
85		Recommended Scoring (5 Total Points)	Consider not giving partial credit since students should understand how to determine range. Look for the misconception of subtracting the first and last given value rather than the maximum and minimum values in the data set.

Question	2b	Benchmark(s)	MA.6.DP.1.2				
90.8		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.				
Question	3	Benchmark(s)	MA.6.DP.1.2				
<table><tr><th>Mode</th><td>19.95</td></tr><tr><th>Interpretation of the Mode</th><td>There are more items on the menu that cost \$19.95 than any other price.</td></tr></table>		Mode	19.95	Interpretation of the Mode	There are more items on the menu that cost \$19.95 than any other price.	Recommended Scoring (10 Total Points)	Consider distributing the points as 5 points for the correct mode and 5 points for the correct interpretation. Consider not giving partial credit for the mode. Partial credit may be considered for a student’s interpretation of the mode if the student correctly notes that the mode is the measure of central tendency and is the value that is found most often in the data set or if a student only notes one of the two pieces of the interpretation of mode. Student responses for the interpretation may be worded differently.
Mode	19.95						
Interpretation of the Mode	There are more items on the menu that cost \$19.95 than any other price.						

Question	4a	Benchmark(s)	MA.6.DP.1.3
	Complete the statements. The values between the box labeled ☐ A ☒ B ☐ C ☐ D ☐ E and the box labeled ☐ A ☐ B ☐ C ☒ D ☐ E represent the middle 50% of the data. The percentage of data between the box labeled C and the box labeled D is ☐ less than ☐ more than ☒ the same as the percentage of data between the box labeled D and the box labeled E.	Recommended Scoring (5 Total Points)	Consider not giving partial credit.
Question	4b	Benchmark(s)	MA.6.DP.1.2
	9.46	Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	4c	Benchmark(s)	MA.6.DP.1.2
	6.67	Recommended Scoring (5 Total Points)	Consider not giving partial credit since students should understand how to determine range. Look for misconception of subtracting the first and last given value rather than the maximum and minimum values in the data set.

Question	5a	Benchmark(s)	MA.6.DP.1.1
	Complete the statement. ○ Erika ○ Jamal ○ Nina ○ Robert ○ Tiana did not write a statistical question because the question ○ is not numerical. ○ is not categorical. ○ does not have variability. <i>Student could also correctly complete the statement saying that Erika did not write a statistical question.</i>	Recommended Scoring (5 Total Points)	Consider not giving partial credit since students should answer all parts of the statement correctly to show understanding.
Question	5b	Benchmark(s)	MA.6.DP.1.1
	Answers will vary. Examples: • How long does it take students to walk to school? • What is the average distance students live from school? • What is the average amount of money students spend on lunch?	Recommended Scoring (5 Total Points)	If a student selects the incorrect person for 1a but the question they wrote for 1b is statistical, the student should not lose any additional points for this part.

Question	6	Benchmark(s)	MA.6.DP.1.3
	Complete the statements. The fastest runner completed the race in ☐ 16.2 ☐ 21.8 ☐ 26.1 minutes. Approximately 50% of the fastest runners completed the race between ☐ 16.2 and 21.4 ☐ 16.2 and 25.2 ☐ 16.2 and 26.1 ☒ 21.4 and 25.2 ☐ 21.4 and 26.1 minutes. The interquartile range is ☐ 3.4 ☒ 3.8 ☐ 5.2 ☐ 9.9 minutes.	Recommended Scoring (15 Total Points)	Consider giving students 5 points for each correctly completed statement. Each statement represents a slightly different concept.
Question	7a	Benchmark(s)	MA.6.DP.1.4
	Complete the statements. The shape of the distribution of received text messages can be best described as ☒ skewed, ☐ uniform, ☐ symmetric, with a ☐ gap ☐ peak ☒ cluster between 40 and 55 text messages. This distribution has a ☒ gap ☐ peak ☐ cluster between 20 and 30 text messages and an outlier between ☐ 30 and 35 ☐ 55 and 100 ☒ 100 and 105 text messages.	Recommended Scoring (20 Total Points)	Consider giving students 5 points for each correctly completed answer box. Each answer box represents a slightly different concept.

Question	7b	Benchmark(s)	MA.6.DP.1.2
17.5		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.